Consider the triangle with vertices:

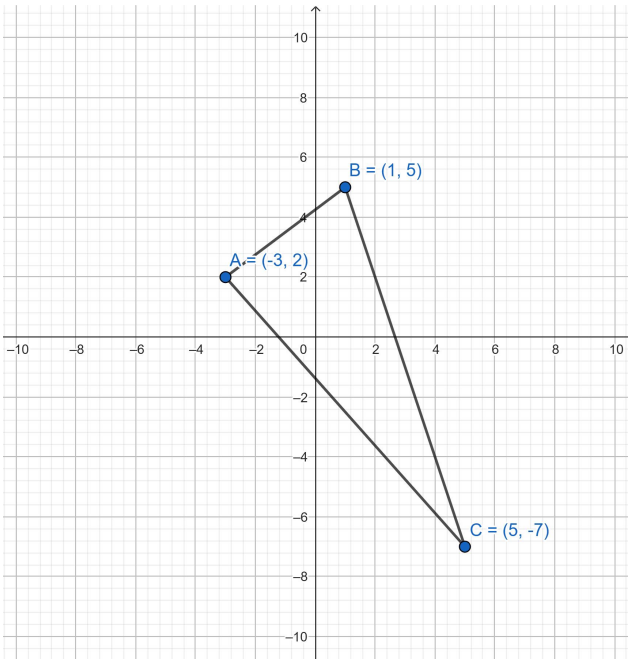
A(−3,2), B(1,5),C(−5,7) (T1)

a. Write down all 8 equations of the side AB
b. Find the perimeter, area, angles, centroid, orthocenter, incenter, and circumcenter of the triangle
c. Write down the equation of median, altitude and angle bisector line dropped from vertex A
d. Evaluate the length of the median, altitude, angle bisector line dropped from the vertex A
e. Write down the parametric and analytic equations of the circumcircle and incircle
f. Evaluate the perimeter and area of the circumcircle and in circle
g. Use 2D-Geogebra and plot the triangle with its incircle and circumcircle

Answer:

Consider the triangle with vertices

A (-3,2), B (1,5), C (5, -7)



Vectors:

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{bmatrix} 1-(-3) \\ 5-2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$|\overrightarrow{AB}| = \sqrt{4^2 + 3^2} = \sqrt{16+9} = \sqrt{25} = 5$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{bmatrix} 5-(-3) \\ -7-2 \end{bmatrix} = \begin{bmatrix} 8 \\ -9 \end{bmatrix}$$

$|\overrightarrow{AC}|=\sqrt{8^2+(-9)^2}=\sqrt{64+81}=\sqrt{145}$

$\overrightarrow{BC}=\overrightarrow{OC}-\overrightarrow{OB}=\begin{bmatrix}5-1\\-7-5\end{bmatrix}=\begin{bmatrix}4\\-12\end{bmatrix}$

$|\overrightarrow{BC}|=\sqrt{4^2+(-12)^2}=\sqrt{16+144}=\sqrt{160}$

A. Write down all 8 equations of the side AB

i. **Vector equation:**

$\overrightarrow{AB}: \vec{r}=\overrightarrow{OA}+\lambda\overrightarrow{AB}$

ii. Parametric equation:

AB: $\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}-3\\2\end{bmatrix}+\lambda\begin{bmatrix}4\\3\end{bmatrix}$

iii. Canonical equation:

$\boldsymbol{AB}: \frac{x+3}{4}=\frac{y-2}{3}$

iv. Analytical implicit equation:

$AB: 3x+9-4y+8=0$

$\boldsymbol{AB: 3x-4y+17=0}$

v. Analytical explicit equation:

$\boldsymbol{AB: y=\frac{3}{4}x+\frac{17}{4}}$

vi. Normal equation:

$\boldsymbol{x\cos\alpha+y\sin\alpha-p=0}$

$\frac{A}{\sqrt{A^2+B^2}}=\boldsymbol{\cos\alpha}$

$\frac{B}{\sqrt{A^2+B^2}}=\boldsymbol{\sin\alpha}$

Where A and B are 3 and -4 respectively

$\boldsymbol{\cos\alpha}=\frac{3}{\sqrt{3^2+4^2}}=\frac{3}{5}$

$\boldsymbol{\sin\alpha}=\frac{-4}{\sqrt{3^2+4^2}}=\frac{-4}{5}$

$p=\frac{C}{\sqrt{A^2+B^2}}=\frac{17}{\sqrt{3^2+4^2}}=\frac{17}{5}$

$p=\frac{17}{5}$

So, the normal equation is

AB: $\frac{3}{5}x-\frac{4}{5}y-\frac{17}{5}=0$

vii. Polar equation:

r=$\frac{p}{Cos(\alpha-\theta)}$

r= $\frac{17}{5Cos(\alpha-\theta)}$

viii. Intercept formula :

AB : $\frac{x}{a}+\frac{y}{b}=1$

X-intercept :$\begin{cases} y=\frac{3}{4}x+\frac{17}{4} \\ y=0 \end{cases}$ p$\left(\frac{-17}{3},0\right)$

Y-intercept :$\begin{cases} y=\frac{3}{4}x+\frac{17}{4} \\ x=0 \end{cases}$ p$\left(0,\frac{17}{4}\right)$

AB: $\frac{x}{\frac{-17}{3}}+\frac{y}{\frac{17}{4}}=1$

AB: $-\frac{3x}{17}+\frac{4y}{17}=1$

B. <u>Find the perimeter, area, angles, centroid, orthocenter, incenter and circumcenter of the triangle</u>

i) <u>Perimeter</u>

P_{ABC} = ($\|\overrightarrow{AB}\|+\|\overrightarrow{BC}\|+\|\overrightarrow{AC}\|$) LU

P_{ABC} = (5+4$\sqrt{10}$+$\sqrt{145}$)LU

ii) <u>Area</u>

$S_{ABC}=\frac{1}{2}\begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix}\ squ=\frac{1}{2}\begin{vmatrix} -3 & 1 & 5 & -3 \\ 2 & 5 & -7 & 2 \end{vmatrix}\ Squ$

S_{ABC} = $\frac{1}{2}|-12-48|$**= 30** ***Squ***

iii) <u>Angles</u>

Angle $\widehat{A}$

$\text{Cos}\widehat{A}=\frac{\overrightarrow{AB}\cdot\overrightarrow{AC}}{\|\overrightarrow{AB}\|\|\overrightarrow{AC}\|}=\frac{(4x8)+(3x(-9))}{5\sqrt{145}}=\frac{1}{\sqrt{145}}=\frac{\sqrt{145}}{145}$

$\widehat{A}=$ **Cos**$^{-1}\left(\frac{\sqrt{145}}{145}\right)$

Angle $\widehat{B}$

$\text{Cos}\widehat{B}=\frac{\overrightarrow{BA}\cdot\overrightarrow{BC}}{\|\overrightarrow{BA}\|\|\overrightarrow{BC}\|}=(\frac{(4x(-4))+((-12)x(-3))}{5(4\sqrt{10})}=\frac{20}{20\sqrt{10}}=\frac{\sqrt{10}}{10}$

$\widehat{B}=$ **Cos**$^{-1}\left(\frac{\sqrt{10}}{10}\right)$

Angle $\widehat{C}$

$$\text{Cos}\hat{C} = \frac{\overrightarrow{CB}\cdot\overrightarrow{CA}}{\|\overrightarrow{CB}\|\|\overrightarrow{CA}\|} = \frac{\left((4x(-8))+(12x9)\right)}{(\sqrt{145})\ x\ (4\sqrt{10})} = \frac{140}{20\sqrt{58}} = 7\frac{\sqrt{58}}{58}$$

$$\widehat{C} = \textbf{Cos}^{-1}\left(7\ \frac{\sqrt{58}}{58}\right)$$

iv) Centroid

$$G\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right) = G\left(\frac{-3+1+5}{3}, \frac{2+5-7}{3}\right) = G\left(\frac{3}{3}, \frac{0}{3}\right)$$

$$\mathbf{G}\ (\mathbf{1}, \mathbf{0})$$

v) Incenter

$$I\left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c}\right)$$

$$I\left(\frac{-3\sqrt{160} + \sqrt{145} + 25}{\sqrt{160}+\sqrt{145}+5}, \frac{2\sqrt{160} + 5\sqrt{145} - 35}{\sqrt{160}+\sqrt{145}+5}\right)$$

vi) Orthocenter

$$H\left(\frac{\begin{vmatrix} y_1^2+x_2x_3 & y_1 & 1 \\ y_2^2+x_1x_3 & y_2 & 1 \\ y_3^2+x_1x_2 & y_3 & 1 \end{vmatrix}}{2S_{ABC}}, \frac{\begin{vmatrix} x_1 & x_1^2+y_2y_3 & 1 \\ x_2 & x_2^2+y_1y_3 & 1 \\ x_3 & x_3^2+y_1y_2 & 1 \end{vmatrix}}{2S_{ABC}}\right)$$

$$H\left(\frac{\begin{vmatrix} 4+5 & 2 & 1 \\ 25-15 & 5 & 1 \\ 49-3 & -7 & 1 \end{vmatrix}}{60}, \frac{\begin{vmatrix} -3 & 9-35 & 1 \\ 1 & 1-14 & 1 \\ 5 & 25+10 & 1 \end{vmatrix}}{60}\right) = H\left(\frac{\begin{vmatrix} 9 & 2 & 1 \\ 10 & 5 & 1 \\ 46 & -7 & 1 \end{vmatrix}}{60}, \frac{\begin{vmatrix} -3 & -26 & 1 \\ 1 & -13 & 1 \\ 5 & 35 & 1 \end{vmatrix}}{60}\right)$$

H (-2,$\frac{7}{3}$)

vii) Circumcenter

$$O\left(\frac{\begin{vmatrix} y_1 & x_1^2+y_1^2 & 1 \\ y_2 & x_2^2+y_2^2 & 1 \\ y_3 & x_3^2+y_3^2 & 1 \end{vmatrix}}{4S_{ABC}}, \frac{\begin{vmatrix} x_1^2+y_1^2 & x_1 & 1 \\ x_2^2+y_2^2 & x_2 & 1 \\ x_3^2+y_3^2 & x_3 & 1 \end{vmatrix}}{4S_{ABC}}\right)$$

$$O\left(\frac{\begin{vmatrix} 13 & 2 & 1 \\ 26 & 5 & 1 \\ 74 & -7 & 1 \end{vmatrix}}{120}, \frac{\begin{vmatrix} 13 & -3 & 1 \\ 26 & 1 & 1 \\ 74 & 5 & 1 \end{vmatrix}}{120}\right) = \mathbf{O}\left(\frac{300}{120}, -\frac{140}{120}\right)$$

$$\mathbf{O}\left(\frac{5}{2}, -\frac{7}{6}\right)$$

Euler's formula:

$$\overrightarrow{HG} = 2\overrightarrow{GO}$$

$$\overrightarrow{HG} = \overrightarrow{OG} - \overrightarrow{OH} = \begin{bmatrix} 1-(-2) \\ 0-\frac{7}{3} \end{bmatrix} = \begin{bmatrix} 3 \\ -\frac{7}{3} \end{bmatrix}$$

$$2\overrightarrow{GO} = 2\left(\overrightarrow{OO} - \overrightarrow{OG}\right) = 2\begin{bmatrix} \frac{5}{2}-1 \\ -\frac{7}{6}-0 \end{bmatrix} = 2\begin{bmatrix} \frac{3}{2} \\ -\frac{7}{6} \end{bmatrix} = \begin{bmatrix} 3 \\ -\frac{7}{3} \end{bmatrix}$$

C. Write down the equation of the median, altitude, angle bisector line dropped from A

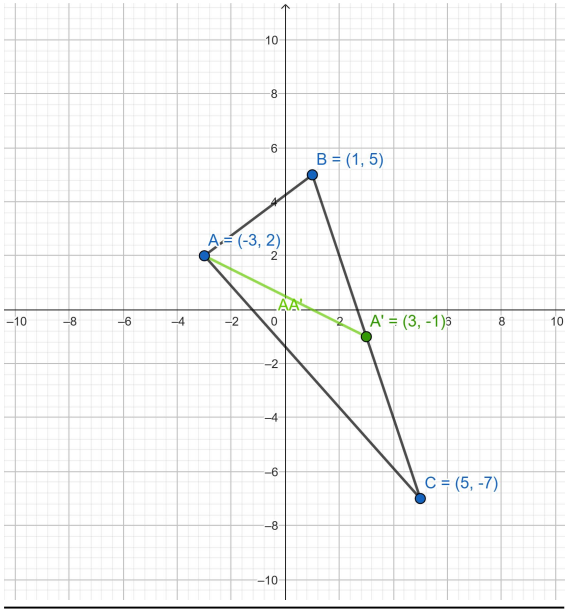
i. Equation of the median dropped from A

A (-3,2)

Let A' be the median dropped from vertex A

A'$\left(\frac{x_C+x_B}{2}, \frac{y_C+y_B}{2}\right)$ = A' $\left(\frac{5+1}{2}, \frac{-7+5}{2}\right)$ = A' (3, -1)

$$\overrightarrow{AA'} = \overrightarrow{OA'} - \overrightarrow{OA} = \begin{bmatrix} 3-(-3) \\ -1-2 \end{bmatrix} = \begin{bmatrix} 6 \\ -3 \end{bmatrix}$$



It's vector equation

$\boldsymbol{m_a}$: $\vec{v}=\overrightarrow{OA}+\lambda\,\overrightarrow{AA'}$

Parametric equation

$\boldsymbol{m_a}$: $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix} + \lambda \begin{bmatrix} 6 \\ -3 \end{bmatrix}$

Canonical equation

$\boldsymbol{m_a}$: $\frac{x+3}{6} = \frac{y-2}{-3}$

Analytical implicit equation:

$\boldsymbol{m_a}$:$-3x - 9 - 6y + 12 = 0$

$\boldsymbol{m_a: 3x + 6y - 3 = 0}$

The length of the median from A

$\boldsymbol{m_a} = \frac{1}{2}\sqrt{2(b^2 + c^2) - a^2}$ LU

$\boldsymbol{m_a} = \frac{1}{2}\sqrt{2(145 + 25) - 160}$ LU

$\boldsymbol{m_a} = \frac{1}{2}\sqrt{180}$ **LU=** $3\sqrt{5}$ **LU**

ii. Equation of altitude dropped from A

$h_a \perp BC : 12x + 4y - 32 = 0$

$h_a \perp BC : y = -3x + 8$

$h_a: y = \frac{1}{3}x + m'$ *since* $A(-3,2)$

$m' = 3$

$h_a : y = \frac{1}{3}x + 3$

h_a: $\boldsymbol{3y - x - 9 = 0}$

iii. Length of altitude from A

$h_a = \frac{2S_{ABC}}{a}\ LU$

$h_a = \frac{2 \times 30}{4\sqrt{10}}\ LU = \boldsymbol{\frac{3\sqrt{10}}{2}\ LU}$

iv. Equation of angle bisector line from A (la)

Equation of AB and AC are $\boldsymbol{3x - 4y + 17 = 0}$ and $\boldsymbol{9x + 8y + 11 = 0}$ respectively

$l_{i,e}$: $\frac{A_1x+B_1y+C_1}{\sqrt{A_1^2+B_1^2}} = \pm \frac{A_2x+B_2y+C_2}{\sqrt{A_2^2+B_2^2}}$

$l_{i,e}$: $\frac{3x-4y+17}{\sqrt{25}} = \pm \frac{9x+8y+11}{\sqrt{145}}$

$l_1 : \frac{3x-4y+17}{5} = \frac{9x+8y+11}{\sqrt{145}}$

$l_1 : 3\sqrt{145}x - 4\sqrt{145}y + 17\sqrt{145} = 45x + 40y + 55$

$l_1 : -4\sqrt{145}y - 40y = -3\sqrt{145}x + 45x + 55 + 17\sqrt{145}$

$l_1 : y(-4\sqrt{145} - 40) = (-3\sqrt{145} + 45)x + 55 + 17\sqrt{145}$

$l_1 : y = (\frac{-3\sqrt{145}+45}{-4\sqrt{145}-40})x + (\frac{55+17\sqrt{145}}{-4\sqrt{145}-40})$ rejected

$l_2 : \frac{3x-4y+17}{5} = -\left(\frac{9x+8y+11}{\sqrt{145}}\right)$

$l_2 : 3\sqrt{145}x - 4\sqrt{145}y + 17\sqrt{145} = -45x - 40y - 55$

$l_2 : y\,(-4\sqrt{145}\ +\ 40)\ =\ -3\sqrt{145}\ -\ 45)\,x\ -\ 55 - 17\sqrt{145}$

$l_2 : y\ =\ (\frac{-3\sqrt{145}-45}{-4\sqrt{145}+40})x\ +\ (\frac{-55-17\sqrt{145}}{-4\sqrt{145}+40})$

Since the gradient of line 2 is positive thus we take the equation of line 2.

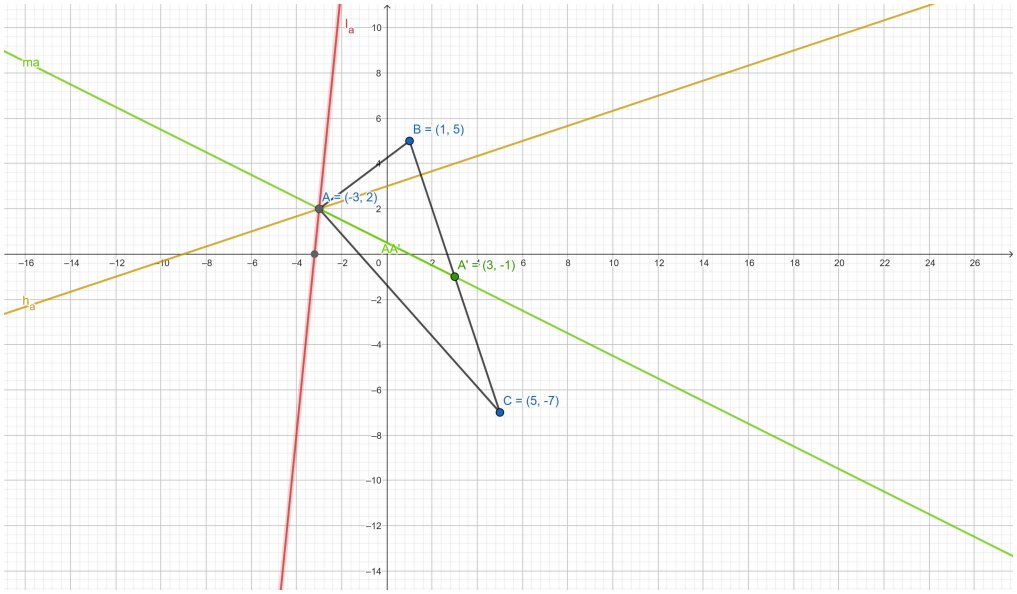
l_a: $(\mathbf{3\sqrt{145}+45})\boldsymbol{x}\ +\ (-\mathbf{4\sqrt{145}}\ +\ \mathbf{40})\,\boldsymbol{y}\ +\ \mathbf{17\sqrt{145}}+\mathbf{55}=\mathbf{0}$ **that is the equation of l_a**

v. The length of l_a

$$l_a\ =\frac{1}{b+c}\sqrt{bc[(b+c)^2-a^2]}\,LU$$

$$l_a\ =\frac{1}{5+\sqrt{145}}\sqrt{5\sqrt{145}\left[\left(5+\sqrt{145}\right)^2-\left(\sqrt{160}\right)^2\right]}\,LU$$

Graph showing the median, altitude and angel bisector line dropped from vertex A.



D. Write down the parametric and analytical equation of the circumcircle and incircle

Parametric Equation of incircle:

$$\begin{cases} x = \mathrm{I}_x + \mathrm{rcost} \\ y = \mathrm{I}_y + \mathrm{rsint} \end{cases}$$

r=$\frac{\mathbf{2}\,S_{ABC}}{P_{ABC}}\,LU$ where r is the inradius

$$\boldsymbol{r}\ =\frac{\mathbf{60}}{\mathbf{5+4\sqrt{10}+\sqrt{145}}}\ \mathbf{LU}$$

$$\Gamma:\begin{cases} \mathbf{x}=\frac{-\mathbf{3\sqrt{160}}\ \ +\mathbf{\sqrt{145}}\ \ +\mathbf{25}}{\mathbf{\sqrt{160}+\sqrt{145}+5}}+\mathbf{2cost} \\ \mathbf{y}=\frac{\mathbf{2\sqrt{160}}\ \ +\mathbf{5\sqrt{145}}\ \ -\mathbf{35}}{\mathbf{\sqrt{160}+\sqrt{145}+5}}+\mathbf{2sint} \end{cases}$$

Analytical Equation of incircle:

$\Gamma: (x - \mathrm{I}_x)^2 + (y - \mathrm{I}_y)^2 = R^2$

$\boldsymbol{\Gamma}: \left(\boldsymbol{x} - \frac{\boldsymbol{-3\sqrt{160}\ \ +\sqrt{145}\ \ +25}}{\boldsymbol{\sqrt{160}+\sqrt{145}+5}}\right)^{\boldsymbol{2}} + \left(\boldsymbol{y} - \frac{\boldsymbol{2\sqrt{160}\ \ +5\sqrt{145}\ \ -35}}{\boldsymbol{\sqrt{160}+\sqrt{145}+5}}\right)^{\boldsymbol{2}} = \left(\frac{\boldsymbol{60}}{\boldsymbol{5+4\sqrt{10}+\sqrt{145}}}\right)^{\boldsymbol{2}}$

Parametric Equation of circumcircle:

$\Gamma: \begin{cases} \mathrm{x} = O_x + \mathrm{Rcost} \\ \mathrm{y} = O_y + \mathrm{Rsint} \end{cases}$

$\boldsymbol{R} = \frac{\boldsymbol{abc}}{\boldsymbol{4S_{ABC}}}$ where R is the circumradius

$$\boldsymbol{R} = \frac{5 \times 4\sqrt{10} \times \sqrt{145}}{\boldsymbol{120}} = \frac{\boldsymbol{20\sqrt{1450}}}{\boldsymbol{120}} = \frac{\boldsymbol{5\sqrt{58}}}{\boldsymbol{6}}\ \boldsymbol{LU}$$

$\begin{cases} \mathrm{x} = \frac{\boldsymbol{-5}}{\boldsymbol{2}} + \frac{\boldsymbol{5\sqrt{58}}}{\boldsymbol{6}}\mathrm{cost} \\ \mathrm{y} = \frac{\boldsymbol{7}}{\boldsymbol{6}} + \frac{\boldsymbol{5\sqrt{58}}}{\boldsymbol{6}}\mathrm{sint} \end{cases}$ where t is the parameter, and it varies from 0 to 2π

Analytical Equation of circumcircle:

$\Gamma: (x - O_x)^2 + (y - O_y)^2 = R^2$

$\boldsymbol{\Gamma}: \left(\mathbf{x} - \frac{\boldsymbol{5}}{\boldsymbol{2}}\right)^{\boldsymbol{2}} + \left(\mathbf{y} + \frac{\boldsymbol{7}}{\boldsymbol{6}}\right)^{\boldsymbol{2}} = \left(\frac{\boldsymbol{5\sqrt{58}}}{\boldsymbol{6}}\right)^{\boldsymbol{2}}$

E. Evaluate the perimeter and the Area of circumcircle and incircle:

The perimeter of incircle

$P_\gamma = 2r\pi\ LU$

$$P_\gamma = 2 \times \frac{\boldsymbol{60}}{\boldsymbol{5 + 4\sqrt{10} + \sqrt{145}}}\pi\ LU = \frac{\boldsymbol{120\pi}}{\boldsymbol{5 + 4\sqrt{10} + \sqrt{145}}}\ \boldsymbol{LU}$$

The Area of incircle

$A_\gamma = \pi\, r^2 square\ units$

$$A_\gamma = \boldsymbol{\pi}\left(\frac{\boldsymbol{60}}{\boldsymbol{5 + 4\sqrt{10} + \sqrt{145}}}\right)^{\boldsymbol{2}}\ \boldsymbol{squ}$$

The perimeter of circumcircle

$P_\Gamma = 2R\pi\ LU$

$$P_\Gamma = 2 \times \frac{\boldsymbol{5\sqrt{58}}}{\boldsymbol{6}}\pi\ LU$$

$$P_\Gamma = \frac{\boldsymbol{5\pi\sqrt{58}}}{\boldsymbol{3}}\ \boldsymbol{LU}$$

The Area of circumcircle

A_Γ=π R^2 Squ

A_Γ**=π** $\left(\frac{\boldsymbol{5\sqrt{58}}}{\boldsymbol{6}}\right)^{\boldsymbol{2}}$ **Squ**

F. Use 2-D GeoGebra and plot the triangle with its incircle and circumcircle

$\Gamma: \left(x - \frac{-3\sqrt{160} + \sqrt{145} + 25}{\sqrt{160}+\sqrt{145}+5}\right)^2 + \left(y - \frac{2\sqrt{160} + 5\sqrt{145} - 35}{\sqrt{160}+\sqrt{145}+5}\right)^2 = \left(\frac{60}{5+4\sqrt{10}+\sqrt{145}}\right)^2$ **Equation of incircle**

$\Gamma: \left(x - \frac{5}{2}\right)^2 + \left(y + \frac{7}{6}\right)^2 = \left(\frac{5\sqrt{58}}{6}\right)^2$ **Equation of circumcircle**

